

e.g., York University  
e.g., Lassonde School of Engineering  
e.g., Earth & Space Science & Engineering

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## Assignment Title

---

Course Code: Course Name

*Student:*

Your Name

123456789

email@my.yorku.ca

*Instructor:*

Dr. Professor

Due date

## Abstract

This document is a beginner-friendly guide to this template. Each section introduces one or more packages included in the template, explains what it does in plain terms, and shows a simple working example. To get the most out of this file, open the compiled PDF and the source `example.tex` file side by side. Reading the code while looking at the result it produces is the fastest way to understand how each feature works.

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# 1 Conditionals

## 1.1 xifthen

This package is used internally by the template and is not typically invoked directly by the user.

The `xifthen` package lets the template make simple yes/no decisions based on your metadata settings in `main.tex`. For example, it checks whether `\metaShowTodo` is set to `true` and shows or hides the todo list accordingly. You do not need to write any `xifthen` commands yourself, just set your metadata values and the template handles the rest.

## 1.2 xstring

This package is used internally by the template for the smart header logic. Users do not use it directly.

The `xstring` package gives the template tools to read and work with text strings. The template uses it to check the length of `\metaCourseName` and decide whether to show the full name or a shorter acronym in the header. The table below shows what `xstring` can do:

**Table 1.1:** Common `xstring` operations.

| Operation    | Command                                     | Result   |
|--------------|---------------------------------------------|----------|
| Extraction   | <code>\StrLeft{Lassonde}{4}</code>          | Lass     |
| Substitution | <code>\StrSubstitute{12/05/26}{/}{-}</code> | 12-05-26 |
| Length       | <code>\StrLen{YorkU}</code>                 | 5        |
| Counting     | <code>\StrCount{Engineering}{n}</code>      | 3        |

# 2 Colours

## 2.1 xcolor

This package is used internally by the template for colour definitions and tints. Users may use it for custom colouring, but it is not required for normal document preparation.

The `xcolor` package adds colour support to  $\text{\LaTeX}$ . The template uses it to apply your chosen theme colour to headings, horizontal rules, boxes, and highlighted notes throughout the document.

### 2.1.1 Setting the Theme Colour

You set the theme colour by editing `\metaThemeColour` in `main.tex`. You can type a hex colour code, the name of any built-in colour, or the name of a custom colour defined in the template. If you leave it empty, the template uses `DefaultBlue`.

**Table 2.1:** Accepted values for `\metaThemeColour`.

| Type          | Example Value             | Effect                               |
|---------------|---------------------------|--------------------------------------|
| Empty         |                           | Defaults to <code>DefaultBlue</code> |
| Hex code      | <code>#810001</code>      | Parsed as an HTML hex colour         |
| Named colour  | <code>red</code>          | Resolved via <code>\colorlet</code>  |
| Custom colour | <code>LassondeNavy</code> | Any colour defined in the sty file   |

The same applies to `\metaLinkColour`, `\metaCiteColour`, and `\metaUrlColour`; each one also falls back to `themecolour` when left empty.

### 2.1.2 Custom Defined Colours

These colours are built into the template and can be typed directly into `\metaThemeColour` or any colour command:

**Table 2.2:** Custom colours defined in `stemtemplate.sty`.

| Name                                         | Hex                  | Preview                                                                              |
|----------------------------------------------|----------------------|--------------------------------------------------------------------------------------|
| <code>DefaultBlue</code> <sup>a</sup>        | <code>#1B4F8A</code> |  |
| <code>YorkRedDark</code>                     | <code>#810001</code> |  |
| <code>ScienceSkyBlueDark</code> <sup>b</sup> | <code>#065C87</code> |  |
| <code>LassondeNavy</code> <sup>c</sup>       | <code>#003366</code> |  |

<sup>a</sup> Unaffiliated professional blue.

<sup>b</sup> Recommended for Faculty of Science students.

<sup>c</sup> Recommended for Lassonde students.

### 2.1.3 Built-in Named Colours

You can also use any of `xcolor`'s built-in colour names. A small selection is shown below. For the full list, see <https://mirrors.ctan.org/macros/latex/contrib/xcolor/xcolor.pdf>:

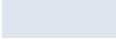
**Table 2.3:** Built-in xcolor named colours.

| Name                                                         | Preview                                                                           | Name        | Preview                                                                             |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------|
| <i>Base colours</i>                                          |                                                                                   |             |                                                                                     |
| red                                                          |  | cyan        |  |
| green                                                        |  | magenta     |  |
| blue                                                         |  | yellow      |  |
| black                                                        |  | white       |  |
| gray                                                         |  | brown       |  |
| <i>Extended colour sets (dvipsnames, svgnames, x11names)</i> |                                                                                   |             |                                                                                     |
| NavyBlue                                                     |  | ForestGreen |  |
| Maroon                                                       |  | BurntOrange |  |
| RoyalPurple                                                  |  | Sepia       |  |
| SlateGray                                                    |  | Teal        |  |
| SteelBlue4                                                   |  | Firebrick3  |  |

#### 2.1.4 Theme Colour and Tints

Adding ! followed by a number after any colour name creates a lighter version of that colour, called a tint. The number is the percentage of the original colour, so `themecolour!70` is 70% colour and 30% white. You can use any percentage directly inside any colour command:

**Table 2.4:** Active theme colour and tints.

| Name                        | Tint | Preview                                                                              |
|-----------------------------|------|--------------------------------------------------------------------------------------|
| <code>themecolour</code>    | 100% |  |
| <code>themecolour!70</code> | 70%  |  |
| <code>themecolour!40</code> | 40%  |  |
| <code>themecolour!15</code> | 15%  |  |

#### 2.1.5 Using `\textcolor` and `\colorbox`

Use `\textcolor` to change the colour of a word or phrase in your text. It takes two arguments: the colour name and the text to colour. For example, `\textcolor{themecolour}{Sample Text}` colours the text without changing anything else on the line.

Use `\colorbox` to place a coloured background behind text, useful for drawing atten-

tion to a short note or label. For example, `\colorbox{themecolour!15}{Sample Text}` produces a lightly shaded highlight box.

## 3 Page Geometry

### 3.1 geometry

This package is used internally by the template to configure page margins and layout. Users do not normally modify these settings.

The geometry package controls the page margins and the size of the text area. The template uses it to set standard margins automatically, you do not need to touch anything for your document to be formatted correctly.

### 3.2 setspace

This package is used internally by the template to control line spacing. Users normally do not interact with it directly, but spacing can be adjusted through metadata commands in `main.tex`.

The setspace package controls the space between lines of text. You can choose between single, one-and-a-half, or double spacing by setting the appropriate metadata value in `main.tex`, you do not need to call any setspace commands yourself.

### 3.3 pdfscape

This package is available for rotating wide tables or figures into landscape orientation. It is not required for normal document preparation and is only used when the author explicitly includes a landscape environment.

The pdfscape package lets you rotate a page sideways so that wide content fits. Wrap anything in a landscape environment and that page will display in landscape orientation in the PDF viewer. The page after it returns to portrait automatically.

The table below is an example of content that is too wide for portrait and benefits from a landscape page:

**Table 3.1:** Wide data table spanning landscape page.

| Parameter        | Trial 1 | Trial 2 | Trial 3 | Trial 4 | Trial 5 |
|------------------|---------|---------|---------|---------|---------|
| Temperature (°C) | 23.4    | 24.1    | 22.8    | 23.9    | 24.5    |
| Pressure (kPa)   | 101.3   | 101.1   | 101.5   | 100.9   | 101.4   |
| Humidity (%)     | 48.7    | 50.2    | 47.9    | 51.0    | 49.3    |
| Voltage (V)      | 4.98    | 5.01    | 4.97    | 5.02    | 5.00    |
| Current (mA)     | 120.1   | 119.8   | 120.4   | 120.0   | 119.6   |

Use `\thispagestyle{plain}` on the line after `\begin{landscape}` to remove the header from the rotated page. Use `\thispagestyle{empty}` instead if you also want to hide the page number in the footer.

## 4 Language & Font

### 4.1 babel

This package manages language-specific typographic rules. The template loads it automatically with English as the active language.

The `babel` package tells  $\text{\LaTeX}$  which language the document is written in. The template loads it with English so that words are hyphenated correctly and punctuation follows English conventions. No changes are needed on your end.

### 4.2 fontspec

This package provides system font access under XeLaTeX. It is loaded automatically by the template.

The `fontspec` package is the foundation of font loading under Xe $\text{\LaTeX}$ . It allows the template to load any font installed on your system by name, and handles all character encoding internally. You do not need to configure it; the template applies it automatically based on your metadata.

#### 4.2.1 `\metaSerifFont` and `\seriftext`

The serif font is configured through metadata. Users typically only use the `\seriftext{}` command directly.

The `\metaSerifFont` metadata key sets the serif font for the document. Leave it empty to use the default, IBM Plex Serif. To override it, set it to any font installed on your system by exact name, for example `Georgia` or `Times New Roman`.

Use `\seriftext{}` to apply the serif font to a specific word or phrase regardless of which font the document body is using:

This text is rendered in the serif font.

#### 4.2.2 `\metaSansFont` and `\sansseriftext`

The sans-serif font is configured through metadata. Users normally interact with it only through `\sansseriftext{}`.

The `\metaSansFont` metadata key sets the sans-serif font for the document. Leave it empty to use the default, IBM Plex Sans. To override it, set it to any font installed on your system by exact name, for example `Arial` or `Helvetica Neue`.

Use `\sansseriftext{}` to apply the sans-serif font to a specific word or phrase:

This text is rendered in the sans-serif font.

### 4.2.3 Default Font

The template selects the global font through the metadata in `main.tex`; users do not modify font packages directly.

You control which font family the document body uses by setting `\metaDefaultFont` in `main.tex`. The current setting is `sans`. Set it to `sans` for the sans-serif font or `serif` for the serif font. The local commands `\seriftext{}` and `\sansseriftext{}` always work regardless of which global font is active.

To use a required font such as Arial for an assignment, set `\metaSansFont{Arial}` and `\metaDefaultFont{sans}`. The serif font remains available via `\seriftext{}` as a companion.

## 5 Spacing Around Headings

### 5.1 `titlesec`

This package is used by the template to control heading formatting and spacing. Users typically do not modify it directly.

The `titlesec` package controls how headings look throughout the document. The template uses it to make all headings bold and coloured with `themecolour`. You do not need to change anything here.

#### 5.1.1 Abstract environment

The abstract environment is customised by the template to ensure consistent formatting. Users typically write their abstract normally without modifying this configuration.

By default,  $\text{\LaTeX}$  centres the abstract text in italic. The template overrides this so the abstract looks like a regular section heading instead, which is the bold flush-left “Abstract” heading you see at the start of this document. You write your abstract the same way either way, just inside `\begin{abstract}` and `\end{abstract}`.

The title page uses a separate version of the abstract that keeps the original centred style, so both layouts are available depending on where the abstract appears.

## 6 Math

### 6.1 `amsmath`, `amssymb`, `amsfonts`

These three packages are the standard tools for writing math in  $\text{\LaTeX}$ . Together they give you display math environments, a large collection of math symbols, and blackboard



bold letters for number sets like  $\mathbb{R}$ ,  $\mathbb{Z}$ , and  $\mathbb{N}$ .

### 6.1.1 equation

The `equation` environment puts a single equation on its own line with a number on the right. Use `equation*` if you do not want the number:

$$2 + 2 = 4 \tag{6.1}$$

$$\boxed{a^2 + b^2 = c^2} \tag{6.2}$$

### 6.1.2 align

The `align` environment lets you write multiple equations that line up at the `&` character. Each line gets its own number unless you add `\nonumber` to that line:

$$\begin{aligned} y &= 2x + 1 \\ &= 2(3) + 1 \end{aligned} \tag{6.3}$$

### 6.1.3 gather

The `gather` environment puts each equation on its own centred line with no alignment between them. Use it when you have a group of separate equations to display together:

$$x + y = 5 \tag{6.4}$$

$$x - y = 1 \tag{6.5}$$

### 6.1.4 multiline

The `multiline` environment is for a single equation that is too long to fit on one line. The first line is left-aligned, the last line is right-aligned, and any lines in between are centred:

$$\begin{aligned} (a_1 + a_2 + a_3) \times (b_1 + b_2 + b_3) = \\ a_1b_1 + a_1b_2 + a_1b_3 + a_2b_1 + a_2b_2 + \\ a_2b_3 + a_3b_1 + a_3b_2 + a_3b_3 \end{aligned} \tag{6.6}$$

### 6.1.5 flalign

The `flalign` environment works like `align` but pushes the columns to the far left and far right of the page. It is useful when you want one equation flush left and another flush right on the same line:

$$F = ma$$

$$E = mc^2 \quad (6.7)$$

### 6.1.6 alignat

The `alignat` environment is similar to `align` but lets you control the spacing between columns manually. The required argument is the number of column pairs. Use it when `align` produces too much or too little space between groups:

$$x = 3 \quad y = 7 \quad (6.8)$$

$$x = -1 \quad y = 2 \quad (6.9)$$

### 6.1.7 split

The `split` environment breaks one long equation across multiple lines while keeping a single equation number. It must be placed inside `equation`:

$$\begin{aligned} (a + b)^2 &= a^2 + 2ab \\ &\quad + b^2 \end{aligned} \quad (6.10)$$

### 6.1.8 cases

The `cases` environment is used for piecewise functions. It draws an opening brace and lets you write a different expression for each condition:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases} \quad (6.11)$$

## 6.2 mathtools

The `mathtools` package adds a few extra tools on top of `amsmath`. The most useful for everyday writing is `\coloneqq`, which produces the “defined as equal to” symbol:

$$f(x) := 2x + 3 \quad (6.12)$$

It also provides matrix environments with different bracket styles. Each one works the same way but uses different surrounding symbols:

$$\text{pmatrix: } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \text{bmatrix: } \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{vmatrix: } \begin{vmatrix} a & b \\ c & d \end{vmatrix} \quad \text{Vmatrix: } \begin{Vmatrix} a & b \\ c & d \end{Vmatrix} \quad (6.13)$$

### 6.3 `bm`

The `bm` package gives you the `\bm{}` command, which makes a math symbol bold while keeping it italic. This is a common way to write vectors and matrices in science and engineering, since `\mathbf{}` would make the letter upright:

$$\mathbf{F} = m\mathbf{a} \quad \mathbf{v} = \mathbf{u} + \mathbf{a}t \quad (6.14)$$

### 6.4 `siunitx`

The `siunitx` package formats numbers and units correctly. Use `\SI{value}{unit}` when you have a number and a unit together, and `\si{unit}` when you just need the unit on its own.

**Table 6.1:** SI unit examples using `siunitx`.

| Command                                           | Output                    |
|---------------------------------------------------|---------------------------|
| <code>\SI{9.81}{\metre\per\second\squared}</code> | 9.81 m/s <sup>2</sup>     |
| <code>\SI{3e8}{\metre\per\second}</code>          | 3 × 10 <sup>8</sup> m/s   |
| <code>\SI{100}{\kilo\gram}</code>                 | 100 kg                    |
| <code>\SI{1.6e-19}{\coulomb}</code>               | 1.6 × 10 <sup>-19</sup> C |
| <code>\si{\ohm}</code>                            | Ω                         |
| <code>\si{\kilo\pascal}</code>                    | kPa                       |

Units can also be used inline inside a sentence. For example, gravitational acceleration is 9.81 m/s<sup>2</sup>, and standard atmospheric pressure is 101.325 kPa.

## 7 Chemistry

## 7.1 mhchem

The `mhchem` package lets you write chemical formulas and reactions using the `\ce{}` command. You just type the formula naturally and it handles all the subscripts, superscripts, and charges for you.

You can use it inline inside a sentence: water is  $\text{H}_2\text{O}$ , table salt is  $\text{NaCl}$ , and the sulfate ion is  $\text{SO}_4^{2-}$ .

## 7.2 Reaction Environments

This template provides six numbered reaction environments, one for each math display environment. All of them use a separate reaction counter that is independent of equation numbering, so reactions are labelled R1, R2, R3 and equations are labelled separately. When section numbering is enabled via `\metaNumberEquations`, reactions are also numbered by section in the same way.

If you prefer reactions to be numbered sequentially as 1, 2, 3... rather than R1, R2, R3..., simply use the standard math display environments instead of the dedicated reaction environments; in that case, reactions will share the same counter as regular equations.

Use `\label{}` and `\cref{}` to cross-reference reactions exactly as you would an equation.

### 7.2.1 requation

The `requation` environment is the reaction equivalent of `equation`. Use it for a single reaction on its own line:



Adding `->[\Delta]` puts a delta above the arrow, commonly used to show that heat or a catalyst is applied:



### 7.2.2 `ralign`

The `ralign` environment is the reaction equivalent of `align`. Use it to show a multi-step mechanism where each step lines up at the `&` character:



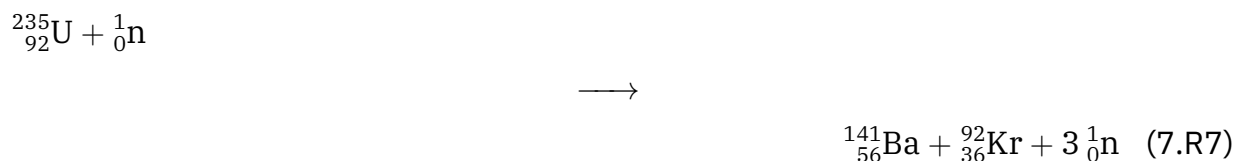
### 7.2.3 `rgather`

The `rgather` environment is the reaction equivalent of `gather`. Use it for a group of related reactions displayed together with no column alignment:



### 7.2.4 `rmultline`

The `rmultline` environment is the reaction equivalent of `multline`. Use it when a single reaction is too long to fit on one line. The first line is left-aligned, the last line is right-aligned, and any lines in between are centred:



### 7.2.5 `rflalign`

The `rflalign` environment is the reaction equivalent of `flalign`. It pushes columns to the far left and far right of the page, useful when you want to display two reactions side by side at opposite ends of the line:



### 7.2.6 `ralignat`

The `ralignat` environment is the reaction equivalent of `alignat`. The required argument is the number of column pairs. Use it when you need precise manual control over the spacing between columns:

$$A + B \longrightarrow C \qquad D \longrightarrow E + F \qquad (7.R9)$$

### 7.2.7 `split`

The `split` environment works exactly the same here as it does in standard math: it breaks a long reaction across multiple lines while keeping a single number. The `split` environment itself does not produce an “R” label, so it must be placed inside `requeation` if you want the reaction to receive an R-style number:

$$\begin{array}{l} A + B \longrightarrow C + D \\ \qquad \qquad \longrightarrow E \end{array} \qquad (7.R10)$$

## 8 Theorem Environments

### 8.1 `amsthm`

The `amsthm` package gives you styled boxes for presenting formal mathematical statements. The template defines three visual styles: `plain`, `definition`, and `remark`. Each style is used by a different group of environments shown below.

#### 8.1.1 Plain — Theorem, Lemma, Proposition, Corollary, Conjecture, Axiom, Proof

**Theorem 8.1:** *For any right triangle with sides  $a$  and  $b$  and hypotenuse  $c$ :  $a^2 + b^2 = c^2$ .*

**Proof:** Draw a square on each side of the triangle and compare the areas. ■

The end-of-proof symbol `\qed` is redefined in `stemtemplate.sty` to show a solid `themecolour` square (■) instead of the default hollow box.

**Lemma 8.1:** *If  $n$  is an even number, then  $n^2$  is also even.*

**Proposition 8.1:** *The sum of two whole-number fractions is always a whole-number fraction.*

**Corollary 8.1:** *If  $p$  is a prime number greater than 2, then  $p$  is odd.*

**Conjecture 8.1:** *Every even number greater than 2 can be written as the sum of two prime numbers.*

**Axiom 8.1:** *Through any two points there is exactly one straight line.*

### 8.1.2 Definition – Definition, Example, Criterion

**Definition 8.1:** An even number is any integer that can be divided by 2 with no remainder.

**Example 8.1:** The numbers 2, 4, 6, and 8 are all even numbers.

**Criterion 8.1:** A number is positive if and only if it is greater than zero.

### 8.1.3 Remark – Remark, Note

**Remark 8.1:** Zero is even because it can be divided by 2 with no remainder.

**Note 8.1:** All theorem environments are automatically numbered by section when `\metaNumberTheorems` is set to true.

## 9 Figures

### 9.1 `graphicx`

The `graphicx` package provides the `\includegraphics` command for placing images in your document. The template is set up to look for images in the `figures/` and `assets/` folders, so you only need to type the filename with no folder path.

You can control the size of the image with optional arguments: `width=0.6\linewidth` scales it to 60% of the text width, `height=5cm` sets a fixed height, and `scale=0.5` makes it half its original size. You only need to set one of these and the other dimension scales automatically to keep the proportions correct.



**Figure 9.1:** Faculty logo included with `\includegraphics`.

### 9.2 `float`

This package is used internally by the template to enable the `[H]` placement specifier. Users interact with it by adding `[H]` to any `figure` or `table` environment.

### 9.3 subcaption

The subcaption package lets you place two or more related figures or tables side by side under a shared caption. Each one gets its own label and caption, and the group gets a shared parent label. They are lettered automatically as (a), (b), (c), and so on.



(a) First sub-figure.



(b) Second sub-figure.

**Figure 9.2:** Two sub-figures using subcaption.

### 9.4 caption

This package is used internally by the template to define caption formatting. Users do not modify it directly; captions are written normally with `\caption{}`.

The caption package controls how figure and table captions look. The template uses it to make the label bold and coloured in `themecolour`, followed by a colon, with the caption text left-aligned in a slightly smaller font. This applies automatically to every `\caption` in your document with no extra work needed.

### 9.5 pdfpages

The pdfpages package lets you insert pages from another PDF directly into your document using `\includepdf[options]{filename.pdf}`. By default the inserted pages have no header or footer. Pass `pagecommand={\thispagestyle{fancy}}` to bring them back.

**Table 9.1:** Common `\includepdf` options.

| Option                                           | Effect                                           |
|--------------------------------------------------|--------------------------------------------------|
| <code>pages=-</code>                             | Include all pages                                |
| <code>pages={1,3}</code>                         | Include pages 1 and 3 only                       |
| <code>pages={2-5}</code>                         | Include a page range                             |
| <code>landscape</code>                           | Rotate inserted pages to landscape               |
| <code>nup=2x1</code>                             | Place two source pages side by side on one sheet |
| <code>pagecommand={\thispagestyle{fancy}}</code> | Restore header and footer on inserted pages      |

## 10 Tables



## 10.1 booktabs

The booktabs package gives you three horizontal line commands for clean, professional tables: `\toprule` at the top, `\midrule` between the header row and the data, and `\bottomrule` at the bottom. Always use these instead of `\hline`.

**Table 10.1:** Basic table using booktabs.

| Parameter  | Value | Unit     |
|------------|-------|----------|
| Resistance | 10    | $\Omega$ |
| Current    | 2     | A        |
| Voltage    | 20    | V        |

## 10.2 array

This package is used internally by the template and is required by `tabularx` and `multirow`. Users may use its column specifiers directly when fine-grained column control is needed.

The array package lets you apply formatting to an entire column at once. The `>...` syntax goes before the column letter in your column definition and applies a command to every cell in that column. The example below uses it to centre and wrap text in two fixed-width columns. The template defines shorthand column types `P{}`, `Q{}`, and `S{}` as ragged2e-aware equivalents of `p{}` for left, centre, and right alignment respectively:

**Table 10.2:** Fixed-width centred columns using array.

| Left column             | Right column          |
|-------------------------|-----------------------|
| Centred text            | Centred text          |
| Long cell content wraps | Also wraps<br>cleanly |

*Note:* Example of a table note.

## 10.3 tabularx

`tabularx` is used internally by `ltablex` and is not the preferred table environment in this template. Use `ltablex` for all tables. The example below is shown for reference only.

The `tabularx` package adds the `X` column type, which automatically stretches to fill the remaining width of the table. You set the total table width as the first argument (usually `\linewidth`) and any `X` columns share the leftover space equally.

The template defines three ragged2e-aware column types built on X: L (ragged right), C (centred), and R (ragged left). These should be preferred over plain X for columns containing wrapped text, as they allow hyphenation and produce more even line lengths.

**Table 10.3:** Auto-width table using `tabularx` with ragged2e column types.

| Parameter  | Value | Unit     | Description                                      |
|------------|-------|----------|--------------------------------------------------|
| Resistance | 10    | $\Omega$ | Opposition to current flow in a circuit          |
| Current    | 2     | A        | Flow of electric charge through a conductor      |
| Voltage    | 20    | V        | Electric potential difference between two points |

## 10.4 `multirow`

The `multirow` package lets a single cell span more than one row. The command is `\multirow{n}{width}{content}`, where `n` is the number of rows to span and `width` is usually `*` to let  $\text{\LaTeX}$  set it automatically. Leave the cells below the merged cell empty. Adding a `\midrule` between groups helps separate them visually:

**Table 10.5:** Table with multi-row cells using `multirow`.

| Category   | Parameter  | Value       |
|------------|------------|-------------|
| Electrical | Resistance | 10 $\Omega$ |
|            | Current    | 2 A         |
| Mechanical | Force      | 50 N        |
|            | Velocity   | 3 m/s       |

## 10.5 `ltablex`

The `ltablex` package extends `tabularx` to support page breaks. It automatically handles tables that are too long to fit on one page, repeating the header on each new page. The X-based column types (L, C, R) work as normal. The `\endfirsthead`, `\endhead`, and `\endfoot` commands define the repeating header and closing rule. Always use `tabularx`, never `longtable` directly:

**Table 10.6:** Table packages and their purposes, demonstrating `ltablex` with an L column.

| Package                            | Purpose                                                                                                                    |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <code>booktabs</code> <sup>a</sup> | Professional horizontal rules ( <code>\toprule</code> , <code>\midrule</code> , <code>\bottomrule</code> )                 |
| <code>array</code> <sup>b</sup>    | Extended column formatting; required by <code>tabularx</code> and <code>multirow</code>                                    |
| <code>tabularx</code> <sup>c</sup> | Flexible X columns that stretch to fill the table width                                                                    |
| <code>multirow</code> <sup>d</sup> | Cells that span more than one row                                                                                          |
| <code>ltablex</code> <sup>e</sup>  | Extends <code>tabularx</code> with page-breaking; always use <code>tabularx</code> , never <code>longtable</code> directly |

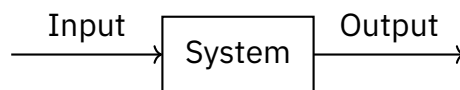
<sup>a</sup> Note on `booktabs`.<sup>b</sup> Note on `array`.<sup>c</sup> Note on `tabularx`.<sup>d</sup> Note on `multirow`.<sup>e</sup> Note on `ltablex`.*Note:* Example of a table note.

## 11 Diagrams & Plots

### 11.1 `tikz`

The `tikz` package lets you draw diagrams directly in your document using simple drawing commands. Everything is described with coordinates and paths in code, so diagrams scale perfectly and can include  $\text{\LaTeX}$  math and text naturally.

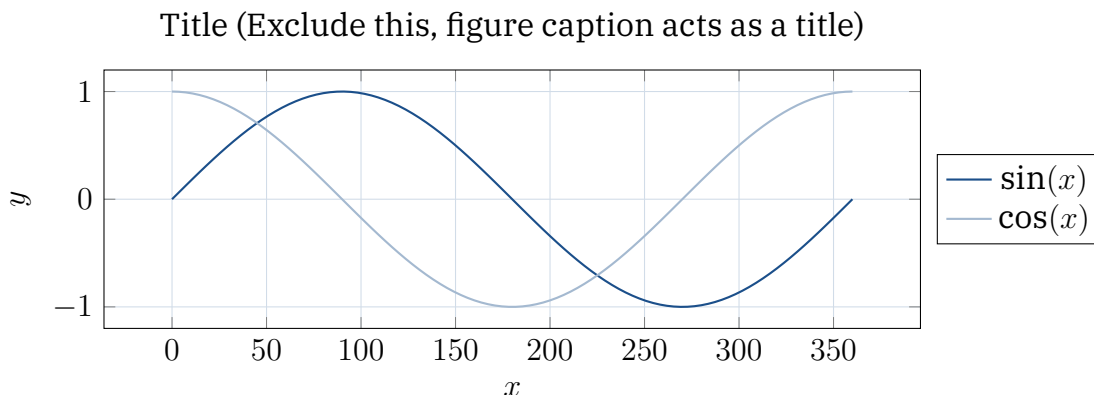
The basic drawing command is `\draw[options] (start) -- (end);`. The `->` option adds an arrowhead and `rectangle` draws a box between two corner coordinates. Coordinates are written as  $(x,y)$  in centimetres by default.

**Figure 11.1:** Simple block diagram drawn with `tikz`.

### 11.2 `pgfplots`

The `pgfplots` package builds data plots on top of `tikz`. Plots go inside an `axis` environment, which handles the axis labels, tick marks, grid lines, and scaling for you. Use `\addplot{expression}` to plot a mathematical function, or supply explicit coordinates to plot data points.

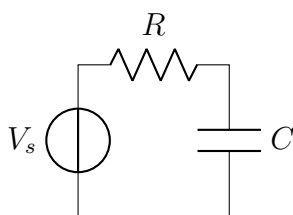
Useful axis options include `width`, `height`, `xlabel`, `ylabel`, `xmin/xmax`, `ymin/ymax`, and `grid`. The `domain` option on `\addplot` sets the range of  $x$  values and `samples` controls how smooth the curve looks.



**Figure 11.2:** Sine and cosine waves plotted with `pgfplots`.

### 11.3 `circuitikz`

The `circuitikz` package extends `tikz` with circuit symbols for drawing electrical schematics. Components are placed along a path using `to[component, label]`, where the component name is a standard symbol such as `R` (resistor), `C` (capacitor), `L` (inductor), `V` (voltage source), or `I` (current source). Paths are chained together to form a closed loop.



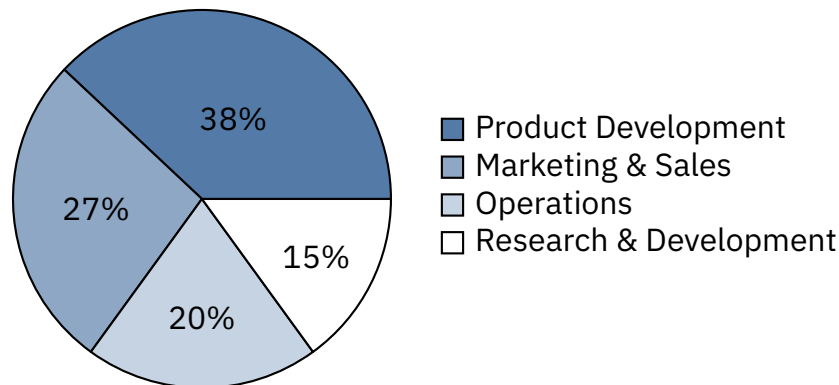
**Figure 11.3:** Simple RC circuit drawn with `circuitikz`.

### 11.4 `pgf-pie`

The `pgf-pie` package draws pie charts directly inside a `tikzpicture` environment. Each slice is specified as a percentage followed by a label, separated by a forward slash, and the values are listed inside the `\pie` command. The package automatically scales the slices so that the percentages add up to a full circle.

Useful options include `radius` to control the size of the chart, `text=legend` to place labels in a legend below instead of on the slices, `text=pin` to connect labels with lines,

and `color` to supply a list of slice colours. Using `themecolour` with tints keeps the chart consistent with the rest of the document.



**Figure 11.4:** Annual budget allocation across departments.

## 12 Algorithms

### 12.1 `algorithm` and `algpseudocode`

The `algorithm` package provides a float wrapper for pseudocode, giving algorithms captions, labels, and an entry in a list of algorithms, the same way `figure` wraps a `tikzpicture`. The `algpseudocode` package supplies the pseudocode commands that go inside the `algorithmic` environment.

The `algorithmic` environment takes an optional line-numbering argument: `[1]` numbers every line, `[2]` numbers every second line, and omitting it disables numbering entirely.

**Table 12.1:** Common `algpseudocode` commands.

| Command                                     | Purpose                                        |
|---------------------------------------------|------------------------------------------------|
| <code>\Require</code>                       | Declares what the algorithm expects as input   |
| <code>\Ensure</code>                        | Declares what the algorithm produces as output |
| <code>\State</code>                         | A single statement on its own line             |
| <code>\Function{name}{params}</code>        | Opens a named function block                   |
| <code>\If / \ElseIf / \Else / \EndIf</code> | Conditional branch                             |
| <code>\For / \EndFor</code>                 | For loop                                       |
| <code>\While / \EndWhile</code>             | While loop                                     |
| <code>\Return</code>                        | Return statement                               |
| <code>\Comment{...}</code>                  | Inline comment, right-aligned                  |

The two algorithms below demonstrate a function with nested control flow. Indentation is handled automatically with no manual spacing required:

---

**Algorithm 12.1:** Binary Search
 

---

**Input:** Sorted array  $A$  of length  $n$ , search value  $target$

**Output:** Index of  $target$  in  $A$ , or  $-1$  if not found

```

1: function BinarySearch( $A, n, target$ )
2:    $low \leftarrow 0$ 
3:    $high \leftarrow n - 1$ 
4:   while  $low \leq high$  do
5:      $mid \leftarrow \lfloor (low + high) / 2 \rfloor$                                 ▷ Compute midpoint
6:     if  $A[mid] = target$  then
7:       return  $mid$                                                     ▷ Target found
8:     else if  $A[mid] < target$  then
9:        $low \leftarrow mid + 1$                                           ▷ Search right half
10:    else
11:       $high \leftarrow mid - 1$                                           ▷ Search left half
12:    end if
13:  end while
14:  return  $-1$                                                          ▷ Target not found
15: end function
  
```

---



---

**Algorithm 12.2:** Bubble Sort
 

---

**Input:** Array  $A$  of length  $n$

**Output:**  $A$  sorted in ascending order

```

1: function BubbleSort( $A, n$ )
2:   for  $i \leftarrow 0$  to  $n - 1$  do
3:     for  $j \leftarrow 0$  to  $n - i - 2$  do
4:       if  $A[j] > A[j + 1]$  then
5:         swap  $A[j]$  and  $A[j + 1]$                                 ▷ Swap out-of-order elements
6:       end if
7:     end for
8:   end for
9: end function
  
```

---

Algorithms are referenced using `\cref` like any other float. Algorithm 12.1 searches a sorted list by repeatedly halving the search range, while Algorithm 12.2 sorts a list by repeatedly swapping neighbouring elements that are in the wrong order.

## 13 Links & References

### 13.1 hyperref

This package is used internally by the template to configure clickable links and PDF meta-data. Users interact with it through `\url{}` and `\href{ }{}` for external links; all internal cross-references are handled automatically.

The `hyperref` package makes every internal cross-reference, citation, and table of contents entry clickable in the PDF. It also embeds document information such as the title and author into the PDF file. The template sets all link colours to `themecolour` automatically.

The two commands you will use directly are `\url{}` for a plain clickable link and `\href{url}{text}` for a link with custom display text:

- `\url{https://www.yorku.ca}` → <https://www.yorku.ca>
- `\href{https://www.yorku.ca}{York University}` → [York University](https://www.yorku.ca)

### 13.2 cleveref

This package is used internally by the template to configure reference formatting. Users interact with it directly through `\cref{}` and `\Cref{}` for all cross-references.

The `cleveref` package replaces the standard `\ref` command with `\cref`, which automatically adds the correct label type in front of the number. For example, `\cref{fig:logo}` produces Fig. 9.1 without you having to type the word “Figure” yourself. Use `\Cref` (capital C) at the start of a sentence. You can also pass multiple labels separated by commas and `cleveref` will group them neatly.

**Table 13.1:** Cross-reference examples using `\cref`.

| Type       | Output         |
|------------|----------------|
| Figure     | Fig. 9.1       |
| Sub-figure | Fig. 9.2a      |
| Table      | Table 10.1     |
| Equation   | Eq. (6.1)      |
| Reaction   | Rxn. (7.R1)    |
| Algorithm  | Algorithm 12.1 |
| Section    | Section 15     |
| Theorem    | Theorem 8.1    |

| Type        | Output                          |
|-------------|---------------------------------|
| Lemma       | Lemma <a href="#">8.1</a>       |
| Proposition | Proposition <a href="#">8.1</a> |
| Corollary   | Corollary <a href="#">8.1</a>   |
| Conjecture  | Conjecture <a href="#">8.1</a>  |
| Axiom       | Axiom <a href="#">8.1</a>       |
| Definition  | Definition <a href="#">8.1</a>  |
| Example     | Example <a href="#">8.1</a>     |
| Criterion   | Criterion <a href="#">8.1</a>   |
| Remark      | Remark <a href="#">8.1</a>      |
| Note        | Note <a href="#">8.1</a>        |
| Problem     | Problem <a href="#">16.1</a>    |
| Listing     | Listing <a href="#">17.1</a>    |

## 14 Typography

### 14.1 `microtype`

This package is used internally by the template to improve text rendering. It requires no user configuration and produces no visually distinct output on its own.

The `microtype` package makes small automatic adjustments to character spacing and sizing to produce cleaner, more even-looking paragraphs. The result is fewer lines that stick out into the margin and less awkward hyphenation, particularly in dense paragraphs with long words. It works silently in the background with no action needed from you.

### 14.2 `xurl`

This package is used internally by the template to improve URL line breaking. Users interact with it through `\url{}`, which is provided by `hyperref`.

The `xurl` package allows long URLs to break across lines at sensible points such as slashes, dots, and hyphens. Without it, a long URL is treated as one unbreakable word and can stick out past the margin. The URL below is long enough to demonstrate a clean line break:

<https://futurestudents.yorku.ca/program/engineering-intl-development-studies>



### 14.3 `indentfirst`

This package is used internally by the template to enforce consistent paragraph indentation. It requires no user configuration.

By default,  $\text{\LaTeX}$  does not indent the first paragraph of a section. The `indentfirst` package changes this so every paragraph is indented, including the first one. The paragraph below is the first in this subsection and is indented because `indentfirst` is loaded:

This is the first paragraph of this subsection, it is indented because `indentfirst` is loaded.

This is the second paragraph, also indented as normal.

### 14.4 `ragged2e`

This package is used internally by the template to improve text alignment in captions, boxes, and tables. Users interact with it through the column types `L`, `C`, `R`, `P{}`, `Q{}`, and `S{}` defined in the template.

The `ragged2e` package provides improved versions of  $\text{\LaTeX}$ 's built-in alignment commands. The standard `\raggedright`, `\centering`, and `\raggedleft` commands disable hyphenation entirely, which can produce very uneven line lengths. The `ragged2e` versions (`\RaggedRight`, `\Centering`, and `\RaggedLeft`) allow hyphenation while still keeping text ragged, resulting in much more even-looking paragraphs. The template applies these automatically to captions, problem and solution boxes, and the custom table column types.

## 15 Lists

### 15.1 `enumitem`

The `enumitem` package gives you extra control over the spacing, labels, and indentation of  $\text{\LaTeX}$ 's three built-in list environments: `itemize`, `enumerate`, and `description`. Options are passed in square brackets directly on the environment.

The most useful option is `noitemsep`, which removes the extra vertical gap that  $\text{\LaTeX}$  normally adds between items. Use it when list items are short and you want a tighter look:

- First item
- Second item
- Third item

The `label` option lets you change the bullet or number style. Use `\arabic*` for num-

bers, `\alph*` for lowercase letters, `\Alph*` for uppercase letters, `\roman*` for lowercase roman numerals, and `\Roman*` for uppercase roman numerals. The `*` is replaced by the item number automatically:

1. First item
2. Second item
3. Third item

- (a) First item
- (b) Second item
- (c) Third item

The `description` environment is useful for glossaries and key-value lists where each item has a bold label followed by a short explanation. The `leftmargin` option controls how far the body text is indented, and `style=nextline` places the body on a new line below the label, which helps when labels are long:

|                             |                                                                                |
|-----------------------------|--------------------------------------------------------------------------------|
| <code>noitemsep</code>      | Removes vertical space between items and between the list and surrounding text |
| <code>label</code>          | Overrides the default bullet or numbering style with any arbitrary format      |
| <code>leftmargin</code>     | Controls the left indent of the list body relative to the page margin          |
| <code>style=nextline</code> | Places the description body on a new line below the label                      |

Lists can be nested up to three levels deep. Indentation and bullet styles adjust automatically at each level:

- First level item
  - Second level item
    - ★ Third level item
- Another first level item

## 16 Problem & Solution Environments

### 16.1 tcolorbox

This package is used internally by the template to define the `problem` and `solution` environments. Users interact with it through those environments directly; the underlying `tcolorbox` configuration requires no modification.

The `tcolorbox` package provides the `problem` and `solution` environments. The `problem` environment is numbered automatically. When `\metaNumberProblems` is set to `true` in `main.tex`, problems are numbered by section (e.g. Problem 2.1, Problem 2.2). When set to `false`, problems are numbered from 1 continuously through the whole document. An optional title can be added in square brackets after `\begin{problem}` and it will appear in the header bar next to the number.

#### Problem 16.1: Ohm's Law

Given a resistor  $R = 10\ \Omega$  with a voltage  $V = 5\ \text{V}$  across it, find the current  $I$ .

#### Solution

Applying Ohm's Law:

$$I = \frac{V}{R} = \frac{5}{10} = 0.5\ \text{A} \quad (16.1)$$



The title argument is optional. When omitted, only the problem number appears in the header:

#### Problem 16.2

A problem without a title is also valid.

## 17 Code Listings

### 17.1 listings

The `listings` package displays source code with syntax highlighting and line numbers. Five named styles are built into this template: `matlab`, `java`, `python`, `c++`, and `latex`. All styles share the same base look: a monospaced font, line numbers on the left in `themecolour`, a light grey background, and automatic line wrapping. Select a style

by adding `style=name` as an option on the `\lstlisting` environment.

For short inline code snippets, use `\lstinline[style=name]|...|` with `|` as the delimiter on both sides.

### 17.1.1 MATLAB

**Listing 17.1:** MATLAB Example.

```
1  x = 0:0.1:2*pi;
2  y = sin(x);
3  plot(x, y);
4  xlabel('x'); ylabel('sin(x)');
5  title('Sine_Wave');
```

### 17.1.2 Java

**Listing 17.2:** Java Example.

```
1  public class HelloWorld {
2      public static void main(String[] args) {
3          System.out.println("Hello, World!");
4      }
5  }
```

### 17.1.3 Python

**Listing 17.3:** Python Example.

```
1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  x = np.linspace(0, 2 * np.pi, 100)
5  y = np.sin(x)
6  plt.plot(x, y)
7  plt.show()
```

### 17.1.4 C++

**Listing 17.4:** C++ Example.

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
```

```
5     cout << "Hello, World!" << endl;
6     return 0;
7 }
```

### 17.1.5 LaTeX

**Listing 17.5:** LaTeX Example.

```
1 \begin{equation}
2     E = mc^2
3     \label{eq:einstein}
4 \end{equation}
5 See \cref{eq:einstein} for Einstein's mass-energy equivalence.
```

## 18 Draft Tools

### 18.1 todonotes

The `todonotes` package lets you leave reminder notes in your document while you are still writing. Notes are controlled by `\metaShowTodo` in `main.tex`. When set to `true` all notes are visible and the todo list appears in the front matter. When set to `false` all notes are hidden globally without you having to delete them from the source.

The default `\todo{}` command places a note in the margin with a line connecting it to the point in the text where it was inserted:

The `inline` option places the note directly in the text flow instead, which is useful when the margin is already crowded:

This is an inline note.

This is a  
margin  
note.

### 18.2 draftwatermark

This package is controlled by `\metaWatermarkText` in `main.tex`. Set the field to any non-empty string to enable the watermark; leave it empty to disable it.

The `draftwatermark` package overlays a large diagonal watermark across every page. You control it entirely through `\metaWatermarkText` in `main.tex`. Set it to any text such as `DRAFT` to show the watermark, or leave it empty to hide it. The watermark is currently disabled. Set `\metaWatermarkText` to a non-empty value in `main.tex` to enable it.

### 18.3 lipsum

This package is used internally by the template for layout testing. It is not intended for use in final documents.

The `lipsum` package generates placeholder text for testing the layout and spacing of a page. Use `\lipsum[n]` to generate paragraph  $n$ , or `\lipsum[1-3]` to generate a range of paragraphs. The text is meaningless Latin and should be replaced with real content before submitting your document.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

## 19 Header & Footer

### 19.1 `fancyhdr`

This package is used internally by the template to configure the page header and footer. Users control the content through metadata fields in `main.tex`; the `fancyhdr` configuration itself requires no modification.

The `fancyhdr` package gives you full control over the header and footer on every page. It divides both the header and footer into three zones: left, centre, and right. Each zone is set with its own command: `\lhead`, `\chead`, and `\rhead` for the header, and `\lfoot`, `\cfoot`, and `\rfoot` for the footer. The template fills these zones automatically using your metadata, so you do not need to call any of these commands yourself.

The decorative rule below the header is coloured in `themecolour` and its thickness is set to 0.8pt by the template. You can also define different headers for different sections of the document or for odd and even pages if needed.

#### 19.1.1 Smart Collision Logic

The template includes a custom `\smartleftheader` command that automatically adjusts the left side of the header to prevent it from overlapping with the right side. It does this by measuring the available horizontal space and choosing one of four display modes:

1. **Full Mode:** Shows the course code followed by the full course name.

2. **Acronym Mode:** If the full name is too long, it is shortened to an acronym (e.g., “Object-Oriented Programming & Telemetry” becomes “OOP&T”).
3. **Minimal Mode:** If even the acronym is too long, only the course code is shown.
4. **Hidden Mode:** If the right side takes up nearly the full width, the left header is hidden completely to keep the header readable.

**Table 19.1:** Dynamic header states based on available width.

| Scenario         | Left Header Output       |
|------------------|--------------------------|
| Not Crowded      | Course Code: Course Name |
| A Little Crowded | Course Code: CN          |
| Crowded          | Course Code              |
| Overcrowded      | (Hidden)                 |

The acronym is generated by the `\makeAcronym` command, which takes the first letter of each significant word in `\metaCourseName` and skips short words like “the”, “of”, and “to”. See Table 1.1 for the string commands it uses internally.

## 20 Bibliography

### 20.1 csquotes

This package is used internally by the template and is required by `biblatex`. It produces no visible output and requires no user configuration.

The `csquotes` package is a behind-the-scenes dependency of `biblatex` that ensures quotation marks in bibliography entries are typeset correctly for the active language. You can also use it directly in your document with `\enquote{}` to wrap a phrase in the correct quotation marks for English: “This is a quoted phrase.”

### 20.2 biblatex

The `biblatex` package handles all citation formatting and bibliography generation. You store your references in a `.bib` file, cite them in the document with `\cite{key}`, and print the full bibliography at the end with `\printbibliography`.

The citation style is set by `\metaBibStyle` in `main.tex`. Examples of supported values are `ieee` and `apa`. The current active style is `ieee`.

After adding new references to your `.bib` file, you need to run Biber once between

compilations for the new citations to resolve. Most editors such as Overleaf and TeXstudio handle this automatically when configured to use Biber as the bibliography backend.

An example citation: [\[1\]](#).



## References

- [1] J. Zhang, K. Fu, J. Qiu, X. Xia, H. Nie, and X. Wen, "Vibration characteristics of CFRP sandwich panels with acoustic black hole structures," *Mechanics of Advanced Materials and Structures*, vol. 33, no. 1, p. 2 445 223, Dec. 26, 2024, issn: 1537-6494. doi: [10.1080/15376494.2024.2445223](https://doi.org/10.1080/15376494.2024.2445223) Accessed: Apr. 26, 2026. [Online]. Available: <https://www.tandfonline.com/doi/full/10.1080/15376494.2024.2445223> cit. on p. 39.

## Appendix A Appendices

Place appendices at the end of your document after the bibliography. The `\appendix` command switches section numbering from arabic numbers to uppercase letters, so each `\section{}` after it becomes a new appendix: Appendix A, Appendix B, and so on. You do not need any extra configuration, just keep adding `\section{}` commands.

**Listing A.1:** Appendix Section Code.

```
1 \appendix
2 \section{First Appendix} % Appendix A
3 ...
4 \section{Second Appendix} % Appendix B
5 ...
```

### A.1 Counter Resets

The template automatically resets the numbering of all major floats at the start of the appendix so they are labelled relative to their appendix letter. For example, the first figure in Appendix B is labelled **Figure B.1**. The following counters are reset automatically:

- figure → A.1, A.2, ...
- table → A.1, A.2, ...
- equation → (A.1), (A.2), ...
- problem → A.1, A.2, ...
- theorem → A.1, A.2, ...
- algorithm → A.1, A.2, ...

### A.2 Section Heading Format

Section headings inside the appendix are automatically formatted as **Appendix A**, **Appendix B**, and so on rather than just **A** or **1**. Subsections and subsubsections are not affected and continue to number normally (A.1, A.1.1, and so on).

### A.3 Cross-Referencing Appendices

Appendix sections, subsections, figures, and tables all support `\label` and `\cref` the same way as the rest of the document. For example, this subsection has the label `sec:appendix-usage-crossref` and can be referenced anywhere in the document like this:

**Listing A.2:** Appendix Section Cross-Reference.

```
1 % In the appendix:  
2 \subsection{Cross-Referencing Appendices}\label{sec:appendix-crossref}  
3  
4 % Elsewhere in the document:  
5 See \cref{sec:appendix-crossref} for details on cross-referencing  
   appendices.
```

Which renders as: See Section [A.3](#) for details on cross-referencing appendices.

## Appendix B Metadata Reference

All document settings are controlled through the metadata block at the top of `main.tex`. Each field is defined with `\newcommand` and read by the template when the document compiles. This appendix lists every available field, what values it accepts, and what it changes in the document.

### B.1 Submission Information

These fields populate the title page and page header. Update them for every new submission.

**Table B.1:** Submission metadata fields.

| Field                             | Description                                                                                                                                                       |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>\metaAssignmentTitle</code> | The assignment or document title shown on the title page.                                                                                                         |
| <code>\metaStudentName</code>     | Your full name. Appears on the title page and in the page header when <code>\metaPartners</code> is empty.                                                        |
| <code>\metaStudentNumber</code>   | Your student number. Shown in brackets after your name in the header when <code>\metaPartners</code> is empty.                                                    |
| <code>\metaStudentEmail</code>    | Your university email address. Shown on the title page.                                                                                                           |
| <code>\metaPartners</code>        | Group partners listed as Name & Number\\Name & Number. When non-empty, the student name and number are removed from the header. Leave empty for solo submissions. |
| <code>\metaUniversity</code>      | University name shown on the title page.                                                                                                                          |
| <code>\metaFaculty</code>         | Faculty name shown on the title page.                                                                                                                             |
| <code>\metaDepartment</code>      | Department name shown on the title page. Leave empty to omit.                                                                                                     |
| <code>\metaCourseCode</code>      | Course code (e.g. EECS 3000). Always shown in the left header.                                                                                                    |
| <code>\metaCourseName</code>      | Full course name. Used by the smart header logic to decide whether to show the full name, a shortened acronym, or nothing depending on available space.           |
| <code>\metaInstructor</code>      | Instructor name shown on the title page.                                                                                                                          |
| <code>\metaDate</code>            | Submission or due date shown on the title page.                                                                                                                   |
| <code>\metaLogoPath</code>        | Path to the faculty or university logo image. Leave empty to omit the logo from the title page.                                                                   |

## B.2 Title Page

**Table B.2:** Title page metadata fields.

| Field                       | Description                                                                                                                                                                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>\metaTitlePage</code> | Controls the title page layout. Accepted values: <code>standard</code> (full title page with all fields), <code>abstract</code> (includes the <code>\metaAbstract</code> field below the title block), <code>compact</code> (condensed single-page layout). |
| <code>\metaAbstract</code>  | Abstract text used when <code>\metaTitlePage</code> is set to <code>abstract</code> . Has no effect for other title page styles.                                                                                                                            |

## B.3 Bibliography

**Table B.3:** Bibliography metadata fields.

| Field                                 | Description                                                                                                                                                                                                            |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>\metaBibFile</code>             | Filename of the <code>.bib</code> reference file (e.g. <code>references.bib</code> ).                                                                                                                                  |
| <code>\metaBibStyle</code>            | Citation and bibliography style. Accepted values: <code>ieee</code> , <code>apa</code> .                                                                                                                               |
| <code>\metaShowBackref</code>         | When <code>true</code> , each bibliography entry shows the page numbers where it was cited. Back-references are coloured using <code>\metaUrlColour</code> . Accepted values: <code>true</code> , <code>false</code> . |
| <code>\metaShowUnusedCitations</code> | When <code>true</code> , all entries in the <code>.bib</code> file appear in the bibliography regardless of whether they were cited. When <code>false</code> , only cited entries appear.                              |

## B.4 Font

**Table B.4:** Font metadata fields.

| Field                       | Description                                                                                                                                                         |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>\metaSerifFont</code> | Serif font loaded by the template. Leave empty for IBM Plex Serif, or set to any system font name (e.g. <code>Georgia</code> ). Used by <code>\seriftext{}</code> . |

| Field                         | Description                                                                                                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>\metaSansFont</code>    | Sans-serif font loaded by the template. Leave empty for IBM Plex Sans, or set to any system font name (e.g. Arial). Used by <code>\sansseriftext{}</code> . |
| <code>\metaDefaultFont</code> | Document body font family. Accepted values: <code>sans</code> (uses <code>\metaSansFont</code> ), <code>serif</code> (uses <code>\metaSerifFont</code> ).   |

## B.5 Misc.

**Table B.5:** Misc. metadata fields.

| Field                            | Description                                                                                                                                      |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>\metaWatermarkText</code>  | Text overlaid as a diagonal watermark on every page. Leave empty to disable. Common values: DRAFT, CONFIDENTIAL.                                 |
| <code>\metaWatermarkScale</code> | Scale factor for the watermark text. Default is 1. Increase for a larger watermark. Has no effect when <code>\metaWatermarkText</code> is empty. |
| <code>\metaLineSpacing</code>    | Line spacing throughout the document. Accepted values: <code>single</code> , <code>onehalf</code> , <code>double</code> .                        |
| <code>\metaShowTodo</code>       | When <code>true</code> , <code>\todo{}</code> notes are visible. When <code>false</code> , all notes are hidden globally.                        |

## B.6 Colours

Each field accepts a hex colour code (with # before it), a built-in  $\text{\LaTeX}$  colour name, or a custom colour defined in the template. Leave any field empty to fall back to the default. See Tables 2.2 and 2.3 for available colour names.

**Table B.6:** Colour metadata fields.

| Field                         | Description                                                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <code>\metaThemeColour</code> | Primary accent colour. Controls headings, rules, boxes, line numbers, and figure labels. Defaults to <code>DefaultBlue</code> when empty. |
| <code>\metaLinkColour</code>  | Colour for internal cross-references and hyperlinks. Leave empty to use <code>themecolour</code> .                                        |
| <code>\metaCiteColour</code>  | Colour for citation references. Leave empty to use <code>themecolour</code> .                                                             |

| Field                       | Description                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------|
| <code>\metaUrlColour</code> | Colour for URLs and bibliography back-references. Leave empty to use <code>themecolour</code> . |

## B.7 Numbering

These fields control whether floats and environments are numbered by section (e.g. Figure 2.1) or continuously through the whole document (e.g. Figure 3). All fields accept true or false.

**Table B.7:** Numbering metadata fields.

| Field                              | Affected Environment     |
|------------------------------------|--------------------------|
| <code>\metaNumberEquations</code>  | equation, reaction       |
| <code>\metaNumberFigures</code>    | figure                   |
| <code>\metaNumberTables</code>     | table                    |
| <code>\metaNumberTheorems</code>   | All theorem environments |
| <code>\metaNumberProblems</code>   | problem                  |
| <code>\metaNumberAlgorithms</code> | algorithm                |
| <code>\metaNumberListings</code>   | listing                  |

## Appendix C Side-by-Side Layouts with `minipage`

The `minipage` environment is a built-in  $\text{\LaTeX}$  tool that creates a self-contained box of a fixed width, like a small independent column on the page. No package is required. It is the most reliable way to place two pieces of content next to each other, especially when one of them contains a display math equation, since equations are not supported inside table cells.

Place two `minipages` side by side by putting `\hfill` between them. `\hfill` pushes the two boxes to opposite sides of the line. Make sure the two widths add up to less than `\linewidth` so there is room for the fill in between.

The optional alignment argument controls how the two `minipages` line up vertically: `[t]` aligns their tops, `[b]` aligns their bottoms, and `[c]` (the default) aligns their centres.

### C.1 Text Side by Side

**Left panel.** This `minipage` occupies 45% of the line width. Content wraps independently within its column and does not affect the right panel.

**Right panel.** `\hfill` between the two `minipages` pushes them to opposite sides. Adjust the widths so they sum to less than `\linewidth` to leave room for the fill.

### C.2 Display Math Side by Side

A common use case is showing two forms of the same expression next to each other for easy comparison:

**Expanded form:**

$$f(x) = x^2 + 2x + 1$$

**Factored form:**

$$f(x) = (x + 1)^2$$

### C.3 Proof Step Comparison

`Minipages` are also useful for showing a simplification step before and after on the same line:

**Before:**

$$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1}$$

**After (for  $x \neq 1$ ):**

$$= x + 1$$

### C.4 Figure and Caption Side by Side

`Minipages` can also place two figures side by side without `subcaption`, when you want each figure to have its own independent caption and label rather than sharing a parent



figure:



**Figure C.1:** First independent figure.



**Figure C.2:** Second independent figure.

Note: \captionof from the caption package is required to place a caption outside a figure float environment.

## Appendix D Notation

This appendix lists the mathematical notation used throughout this document, grouped by subject area. It is an example of how you might document your own notation in a report or assignment.

### D.1 General Mathematics

**Table D.1:** General mathematical notation.

| Symbol                               | Meaning                                                      |
|--------------------------------------|--------------------------------------------------------------|
| $f(x), g(x)$                         | Functions of $x$                                             |
| $f(x) := \dots$                      | Definition of $f(x)$ (defined as equal to)                   |
| $\mathbb{R}, \mathbb{Z}, \mathbb{N}$ | Real numbers, integers, natural numbers                      |
| $\infty$                             | Infinity                                                     |
| $\sum_{n=1}^N$                       | Sum of terms from $n = 1$ to $N$                             |
| $\lfloor x \rfloor$                  | Floor of $x$ , the largest integer less than or equal to $x$ |
| $ x $                                | Absolute value of $x$                                        |

### D.2 Algebra

**Table D.2:** Algebra notation.

| Symbol                                                       | Meaning                                |
|--------------------------------------------------------------|----------------------------------------|
| $a^2 + b^2 = c^2$                                            | Pythagorean theorem                    |
| $(a + b)^2 = a^2 + 2ab + b^2$                                | Binomial expansion                     |
| $ x  = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$ | Absolute value as a piecewise function |
| $\mathbf{v}, \mathbf{u}$                                     | Vectors (bold italic)                  |
| $A, B$                                                       | Matrices (uppercase letters)           |

### D.3 Physics

**Table D.3:** Physics notation.

| Symbol       | Meaning                                 |
|--------------|-----------------------------------------|
| $\mathbf{F}$ | Force vector (N)                        |
| $m$          | Mass (kg)                               |
| $\mathbf{a}$ | Acceleration vector (m/s <sup>2</sup> ) |
| $\mathbf{v}$ | Velocity vector (m/s)                   |
| $t$          | Time (s)                                |
| $V$          | Voltage (V)                             |
| $I$          | Current (A)                             |
| $R$          | Resistance ( $\Omega$ )                 |

## Appendix E Greek Letters Reference

A complete reference of Greek letters available in math mode. Greek letters are used often in science and engineering for variables, constants, and operators.

**Table E.1:** Lowercase Greek letters.

| Symbol        | Command                  | Symbol      | Command                | Symbol     | Command               |
|---------------|--------------------------|-------------|------------------------|------------|-----------------------|
| $\alpha$      | <code>\alpha</code>      | $\iota$     | <code>\iota</code>     | $\rho$     | <code>\rho</code>     |
| $\beta$       | <code>\beta</code>       | $\kappa$    | <code>\kappa</code>    | $\sigma$   | <code>\sigma</code>   |
| $\gamma$      | <code>\gamma</code>      | $\lambda$   | <code>\lambda</code>   | $\tau$     | <code>\tau</code>     |
| $\delta$      | <code>\delta</code>      | $\mu$       | <code>\mu</code>       | $\upsilon$ | <code>\upsilon</code> |
| $\epsilon$    | <code>\epsilon</code>    | $\nu$       | <code>\nu</code>       | $\phi$     | <code>\phi</code>     |
| $\varepsilon$ | <code>\varepsilon</code> | $\xi$       | <code>\xi</code>       | $\varphi$  | <code>\varphi</code>  |
| $\zeta$       | <code>\zeta</code>       | $\pi$       | <code>\pi</code>       | $\chi$     | <code>\chi</code>     |
| $\eta$        | <code>\eta</code>        | $\varpi$    | <code>\varpi</code>    | $\psi$     | <code>\psi</code>     |
| $\theta$      | <code>\theta</code>      | $\varsigma$ | <code>\varsigma</code> | $\omega$   | <code>\omega</code>   |
| $\vartheta$   | <code>\vartheta</code>   |             |                        |            |                       |

**Table E.2:** Uppercase Greek letters (only those that differ from Latin capital letters).

| Symbol   | Command             | Symbol     | Command               | Symbol   | Command             |
|----------|---------------------|------------|-----------------------|----------|---------------------|
| $\Gamma$ | <code>\Gamma</code> | $\Lambda$  | <code>\Lambda</code>  | $\Phi$   | <code>\Phi</code>   |
| $\Delta$ | <code>\Delta</code> | $\Xi$      | <code>\Xi</code>      | $\Psi$   | <code>\Psi</code>   |
| $\Theta$ | <code>\Theta</code> | $\Pi$      | <code>\Pi</code>      | $\Omega$ | <code>\Omega</code> |
| $\Sigma$ | <code>\Sigma</code> | $\Upsilon$ | <code>\Upsilon</code> |          |                     |

## Appendix F Common Math Operators and Symbols

### F.1 Logic and Sets

**Table F.1:** Logic and set notation.

| Symbol            | Command               | Symbol      | Command                |
|-------------------|-----------------------|-------------|------------------------|
| $\Rightarrow$     | <code>\implies</code> | $\in$       | <code>\in</code>       |
| $\Leftrightarrow$ | <code>\iff</code>     | $\notin$    | <code>\notin</code>    |
| $\forall$         | <code>\forall</code>  | $\subset$   | <code>\subset</code>   |
| $\exists$         | <code>\exists</code>  | $\subseteq$ | <code>\subseteq</code> |
| $\neg$            | <code>\neg</code>     | $\cup$      | <code>\cup</code>      |
| $\wedge$          | <code>\land</code>    | $\cap$      | <code>\cap</code>      |
| $\vee$            | <code>\lor</code>     | $\emptyset$ | <code>\emptyset</code> |
| $\oplus$          | <code>\oplus</code>   | $\setminus$ | <code>\setminus</code> |

### F.2 Relations

**Table F.2:** Relation symbols.

| Symbol    | Command              | Symbol    | Command              |
|-----------|----------------------|-----------|----------------------|
| $\equiv$  | <code>\equiv</code>  | $\approx$ | <code>\approx</code> |
| $\cong$   | <code>\cong</code>   | $\sim$    | <code>\sim</code>    |
| $\propto$ | <code>\propto</code> | $\leq$    | <code>\leq</code>    |
| $\neq$    | <code>\neq</code>    | $\geq$    | <code>\geq</code>    |
| $\ll$     | <code>\ll</code>     | $\gg$     | <code>\gg</code>     |

### F.3 Arrows

**Table F.3:** Arrow symbols.

| Symbol            | Command                      | Symbol       | Command                 |
|-------------------|------------------------------|--------------|-------------------------|
| $\rightarrow$     | <code>\rightarrow</code>     | $\leftarrow$ | <code>\leftarrow</code> |
| $\Rightarrow$     | <code>\Rightarrow</code>     | $\Leftarrow$ | <code>\Leftarrow</code> |
| $\Leftrightarrow$ | <code>\Leftrightarrow</code> | $\mapsto$    | <code>\mapsto</code>    |
| $\uparrow$        | <code>\uparrow</code>        | $\downarrow$ | <code>\downarrow</code> |

## Appendix G Font Commands

### G.1 Text Font Shapes and Weights

**Table G.1:** Text font commands.

| Command                      | Output                 | Common use                 |
|------------------------------|------------------------|----------------------------|
| <code>\textbf{...}</code>    | <b>Bold text</b>       | Emphasis, key terms        |
| <code>\textit{...}</code>    | <i>Italic text</i>     | Titles, emphasis           |
| <code>\texttt{...}</code>    | Monospace text         | Code, file names, commands |
| <code>\textsc{...}</code>    | Small Caps             | Acronyms, proper names     |
| <code>\textsf{...}</code>    | Sans-serif text        | Labels, captions           |
| <code>\emph{...}</code>      | <i>Emphasised text</i> | Context-aware italic       |
| <code>\underline{...}</code> | <u>Underlined text</u> | Use sparingly              |

### G.2 Font Sizes

Font size commands are applied by placing them inside a group with curly braces: `{\large This text is large.}`. They scale relative to the base font size of the document.

**Table G.2:** Font size commands.

| Command                    | Output        |
|----------------------------|---------------|
| <code>\tiny</code>         | tiny text     |
| <code>\scriptsize</code>   | script size   |
| <code>\footnotesize</code> | footnote size |
| <code>\small</code>        | small text    |
| <code>\normalsize</code>   | normal text   |
| <code>\large</code>        | large text    |
| <code>\Large</code>        | Large text    |
| <code>\LARGE</code>        | LARGE text    |
| <code>\huge</code>         | huge text     |